CHEST

Do Young Individuals With Chronic E-Cigarette Exposure Display Cardiopulmonary Abnormalities During Exercise and Blunted Recruitment of Pulmonary Diffusing Capacity?

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BACKGROUND: The prevalence of e-cigarette use is increasing, and young adults who report e-cigarette use and no history of tobacco smoking report greater respiratory symptoms. Traditional evaluation of resting pulmonary function may fail to detect subclinical abnormalities.

RESEARCH QUESTION: To what extent do otherwise healthy, young adults who report regular e-cigarette use and no tobacco smoking history exhibit altered cardiopulmonary function during exercise and blunted recruitment of pulmonary diffusing capacity for carbon monoxide (D_{LCO})?

STUDY DESIGN AND METHODS: Twenty participants with chronic e-cigarette exposure (mean age, 23 ± 4 years) with no tobacco smoking history and 20 age-, height-, and sex-matched control participants underwent a pulmonary function test and cardiopulmonary exercise test. Key outcomes included exercise capacity ($\dot{V}O_{2peak}$), ventilatory efficiency (minute ventilation/CO_2 output nadir), exertional dyspnea, and operating lung volumes during exercise. A secondary aim investigated recruitment of resting D_{LCO} and its components, measured as the change from the seated to supine posture. Adjusted linear regression models were used to evaluate the effect of group on key outcome variables at rest and during exercise.

RESULTS: Both groups presented with normal pulmonary function (all, $P_{group} > .05$). Individuals who use e-cigarettes exhibited a lower $\dot{V}O_{2peak}$ ($P_{group} = .017$), elevated minute ventilation/carbon dioxide output nadir ($P_{group} = .037$), and greater exertional dyspnea ($P_{group} < .001$, $P_{group*workload} < .001$), whereas operating lung volumes did not differ from control participants (all, $P > .05$). Individuals who use e-cigarettes also displayed a blunted D_{LCO} recruitment to a postural change ($P_{group*posture} = .036$). Between-group differences in dyspnea and D_{LCO} recruitment were independent of $\dot{V}O_{2peak}$.

INTERPRETATION: In this study, otherwise healthy young adults who report regular e-cigarette use exhibited evidence of abnormal cardiopulmonary responses to exercise and blunted D_{LCO} recruitment. These findings suggest early cardiopulmonary impairment and pulmonary vascular dysfunction in young adults with chronic e-cigarette exposure.

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KEY WORDS: cardiopulmonary exercise testing; diffusing capacity; dyspnea; electronic cigarettes; ventilation

Take-Home Points

Research Question: To what extent do otherwise healthy, young adults who report regular e-cigarette use and no tobacco smoking history exhibit altered cardiopulmonary function during exercise, and altered recruitment of pulmonary diffusing capacity (D_{LCO})?

Results: Participants with chronic e-cigarette exposure exhibited reduced exercise capacity, impaired ventilatory efficiency, exaggerated exertional dyspnea, and reduced recruitment of pulmonary diffusing capacity for carbon monoxide compared with control participants.

Interpretation: The findings of this study suggest early cardiopulmonary impairment and pulmonary vascular dysfunction in young adults with chronic e-cigarette exposure.

E-cigarettes have been widely promoted as a safer alternative to combustible tobacco cigarettes.[1] Public Health England has proposed that e-cigarette use is "95% less harmful than smoking"; however, this assertion has been challenged by the American Thoracic Society Tobacco Action Committee, which contends that it is not supported by sufficient evidence.[2] Moreover, the committee cautions that framing e-cigarette use as "less harmful" may inadvertently normalize or promote uptake. Indeed, the prevalence of habitual e-cigarette use among youth and young adults in the United States, Canada, and England has increased to 6%-25% since 2017,[3] with the prevalence now exceeding that of conventional tobacco smoking in these age groups.[4]

Although evidence regarding the long-term consequences of chronic e-cigarette use is limited, particularly among individuals without prior tobacco smoking exposure, emerging findings indicate that individuals with regular e-cigarette exposure exhibit physiological abnormalities. Young adults with chronic e-cigarette exposure and no tobacco smoking history report greater respiratory symptoms,[5,6] although underlying mechanisms remain unknown. A systematic review showed that spirometry findings are preserved in individuals with e-cigarette exposure.[7] Notably, the review identified elevated airway resistance evaluated according to body plethysmography, suggesting the presence of subclinical abnormalities that may elude detection through routine spirometric assessment. More recent work has shown that young adults with chronic e-cigarette exposure have an impaired lung clearance index,[6] which is an indicator of gas mixing efficiency that has been shown to predict subsequent development of COPD.[8]

Beyond airway function, research has identified pulmonary vascular abnormalities in young adults with chronic e-cigarette exposure and no history of tobacco smoking and otherwise normal pulmonary function.[9] Specifically, young adults with chronic e-cigarette exposure exhibit greater ventilation-perfusion mismatch compared with matched control participants, driven primarily by increased perfusion heterogeneity.[9] Experimental and in vitro work further supports these findings, as both nicotine and the vehicle constituents of e-cigarette aerosols exert deleterious effects on the pulmonary vasculature.[10,11] Because impairments in the pulmonary vasculature have been shown to precede airway damage,[12] a deeper investigation into cardiopulmonary function among young adults with chronic e-cigarette exposure is warranted.

Exercise represents a physiological stressor that provides a unique opportunity to interrogate cardiopulmonary function under dynamic conditions, potentially unmasking abnormalities not evident at rest or during standard clinical testing.[13] To date, only a single study has evaluated the exercise responses of individuals with e-cigarette exposure, reporting a

ABBREVIATIONS: CPET = cardiopulmonary exercise test; D_{LCO} = pulmonary diffusing capacity for carbon monoxide; D_M = pulmonary membrane diffusing capacity; EILV = end-inspiratory lung volume; EELV = end-expiratory lung volume; LME = linear mixed effects; TLC = total lung capacity; $\dot{V}_C$ = pulmonary capillary blood volume; $\dot{V}_E$ = minute ventilation; $\dot{V}CO_2$ = carbon dioxide output; $\dot{V}O_{2peak}$ = peak rate of oxygen consumption

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reduction in peak rate of oxygen consumption ($\dot{V}O_2$peak), an index of exercise capacity.[14] However, that investigation did not include comprehensive assessments of submaximal exercise responses, which could provide mechanistic insight into the reduction in $\dot{V}O_2$peak and the greater respiratory symptoms reported in individuals with e-cigarette exposure.[5,6]

The purpose of the current study was to investigate the cardiopulmonary responses during incremental cycle exercise in young adults with chronic e-cigarette exposure and no history of tobacco smoking. Based on previous work indicating that individuals with e-cigarette exposure exhibit reduced $\dot{V}O_{2peak}$,[14] greater respiratory symptoms,[5,6] and abnormal pulmonary vascular perfusion,[9] it was hypothesized that individuals with e-cigarette exposure would display cardiopulmonary abnormalities, including reduced $\dot{V}O_{2peak}$, impaired ventilatory efficiency (minute ventilation/CO_2 output [$\dot{V}_E/\dot{V}CO_2$] nadir), and exaggerated exertional dyspnea. To further investigate pulmonary vascular function, a secondary purpose was to investigate the recruitment of resting pulmonary diffusing capacity for carbon monoxide (D_{LCO}) and its components, pulmonary capillary blood volume (V_C) and pulmonary membrane diffusing capacity (D_M). Using the postural shift from the seated to supine posture as a hemodynamic stressor,[15-17] we hypothesized that individuals with e-cigarette exposure would display a blunted recruitment of D_{LCO} and V_C compared with control participants.

Study Design and Methods

Participants

A convenience sample of 20 participants with chronic e-cigarette exposure and 20 control participants were recruited from the community between September 2023 and November 2024, and participants provided written informed consent. Control participants were matched to participants with e-cigarette exposure on age, height, and sex. Chronic e-cigarette exposure was defined as daily reported use for $\geq$ 2 years. Control participants were included based on no self-reported history of e-cigarette use. Exclusion criteria for both participants with e-cigarette exposure and control participants included any self-reported history of tobacco smoking ($>$ 1 pack total exposure), significant cannabis exposure ($>$ 1 joint-year total and/or $>$ 4 joint smoking days per week in the past 30 days), and presence of cardiopulmonary disease.

Experimental Overview

This cross-sectional observational study received approval by the Health Research Ethics Board at the University of Alberta (Pro #00119593) and comprised 2 visits. Visit 1 involved written informed consent, a health screening to determine eligibility, a pulmonary function test, and a cardiopulmonary exercise test (CPET). The second visit, separated by a minimum of 48 hours, evaluated D_{LCO}, V_c, and D_M at rest in the seated and supine postures.

Physical Activity Questionnaire and E-Cigarette Exposure History

Participants completed a physical activity questionnaire to evaluate self-reported weekly minutes of moderate to vigorous physical activity.[18] The puff-year metric was used to quantify e-cigarette use as it is designed to equate e-cigarette puffs to tobacco cigarette puffs.[5] This metric calculates an e-cigarette "puff-year" by using self-reported daily inhalation frequency and history of e-cigarette use; it has been shown to be associated with respiratory symptoms in individuals with e-cigarette exposure with no tobacco smoking history.[5]

Pulmonary Function Test and CPET

The pulmonary function test included spirometry, dynamic and static lung volumes, and single-breath D_{LCO} (Encore229 Vmax; SensorMedics). All tests were administered by a single, trained technician to American Thoracic Society/European Respiratory Society Grade A standards per American Thoracic Society/European Respiratory Society 2019 technical standards.[19-21] Pulmonary function test values are reported as absolute values and relative to reference values.[20,22,23]

The incremental CPET was performed on an electronically braked cycle ergometer (Ergoselect 200; ergoline GmbH) and consisted of a 2-minute baseline resting period, followed by stepwise increases in workload of 25 W every 2 minutes, beginning at 50 W, until volitional exhaustion.[13] Standard cardiopulmonary parameters were collected.[13] Ventilatory efficiency ($\dot{V}_E/\dot{V}CO_2$ nadir) was determined as the lowest 30-second value achieved during the CPET.[24-26]

During the final 30 seconds of each 2-minute stage, participants rated the perceived intensity of their breathing

discomfort and leg discomfort using the 0 to 10 modified Borg category ratio scale.[27] Following measurement of perceptual responses at each workload, an inspiratory capacity maneuver was performed to quantify end-inspiratory lung volume (EILV) and end-expiratory lung volume (EELV) as a percentage of total lung capacity (%TLC).[28] Metabolic data from 60 to 90 seconds were averaged for analysis; the final 30 seconds were excluded to avoid influence from the inspiratory capacity maneuver. Exercise capacity ($\dot{V}O_{2peak}$) was reported as absolute and relative values, and relative to reference values.[29] Arterial oxygen saturation was monitored via finger pulse oximetry (Radical 7; Masimo).

D_{LCO}, V_C, and D_M

The second visit measured D_{LCO} and its components, V_C and D_M.[30] Measurements occurred at rest in the upright and supine postures (order randomized) in 16 participants with e-cigarette exposure and 17 control participants. Consistent with previous work,[30] D_{LCO}, V_C, and D_M were assessed by using modified 6-second breath holds at 3 different F_{IO_2} tensions (randomized, 0.21, 0.40, and 0.60), following the Roughton and Forster multiple-F_{IO_2} breath-hold technique.[15,31] D_{LCO} measurements were completed after 10 minutes of rest in each posture. D_{LCO} was adjusted to hemoglobin concentration[23] and alveolar volume.[15,16] Alveolar volume was determined by using methane (0.3%).[15]

Due to the disparity between alveolar volume in the upright and supine positions, all D_{LCO} values were adjusted to alveolar volume as per the previously published equation.[15] The minimum acceptable R^2 value from the D_{LCO} regression equation created by the different concentrations of oxygen was set to 0.96, and D_{LCO} maneuvers were repeated if R^2 values fell below this threshold. Previous work indicates that e-cigarette use does not increase carboxyhemoglobin concentration; therefore, no adjustment was performed.[32] Blood gas analyses among a subset of participants with e-cigarette exposure confirmed that carboxyhemoglobin levels were within normal range (mean carboxyhemoglobin, 1.3% ± 0.2%).

Outcome Measures

Primary analysis investigated maximal and submaximal exercise responses. Single-point primary outcomes were $\dot{V}O_{2peak}$ and $\dot{V}_E/\dot{V}CO_2$ nadir (an index of ventilatory efficiency). Key submaximal exercise responses were exertional dyspnea and operating lung volumes. Secondary outcomes were the recruitment (Δ) of D_{LCO}, Vc, and D_M from the seated to supine posture.

Statistical Analysis

The study sample size was calculated based on previous work showing a lower $\dot{V}O_{2peak}$ in individuals with e-cigarette exposure compared with control participants (mean difference, 4.7 ± 5.0 mL/kg per minute).[14] Assuming a similar between-group difference, 19 participants per group would provide sufficient power to detect a difference in $\dot{V}O_{2peak}$ ($\alpha = .05$; $\beta = 0.20$).

Descriptive statistics are presented as mean ± SD and were analyzed by using 1-way analysis of variance to assess the main effect of group. Normality was assessed by using the Shapiro-Wilk test. Variables found to be non-normally distributed were analyzed by using Mann-Whitney U tests and are reported as median (interquartile range).

Key exercise and pulmonary vascular outcomes are presented as estimated marginal means ± SE, unless otherwise specified. Between-group comparisons of key single-point outcomes were analyzed via linear regression adjusted for age (years), height (centimeters), and sex (male, female), as these variables are well-established predictors of cardiopulmonary function.[22]

Cardiopulmonary responses to submaximal exercise were analyzed by using random intercept linear mixed effects (LME) models. LME models included tests for group, workload, and their interaction as fixed effects. Data up to the highest equivalent work rate (100 W) were included in analyses, and workload was replaced with $\dot{V}_E$ where indicated. Peak exercising data were excluded from the LME models because of variability in peak workloads achieved between participants and to avoid biasing estimates. The repeated effect covariance structure was unstructured, accounting for within-subject correlation for repeated measurements, with random intercepts accounting for between-subject variability. Models were adjusted for age, height, sex, and $\dot{V}O_{2peak}$.

LME models were similarly used to analyze postural recruitment of D_{LCO} and its components. Models included group, posture, and their interaction as fixed effects, and adjusted for age, height, sex, and $\dot{V}O_{2peak}$.[22,29] Unadjusted results are reported in the supplementary files.

Statistical significance was set a priori at $P < .05$. Data were analyzed by using IBM SPSS Statistics version 29 (IBM SPSS Statistics, IBM Corporation).

Results

Participant Characteristics, Pulmonary Function, and E-Cigarette History

Participant characteristics, e-cigarette use history, and pulmonary function are reported in Table 1. Participants with e-cigarette exposure did not differ from control participants in anthropometric measures or self-reported weekly minutes of moderate to vigorous physical activity (all, $P_{group} > .05$). All participants exhibited normal spirometry findings, lung volumes, and resting D_{LCO}, with no between-group differences observed (all, $P_{group} > .05$). Of note, all variables showed normality with the exception of the ratio of FEV_1/FVC, which was accordingly reported as mean (interquartile range) in Table 1. Participants with e-cigarette exposure had a mean e-cigarette exposure history of 3.4 ± 2.9 puff-years, whereas, by design, control participants reported no history of e-cigarette use. All participants with e-cigarette exposure reported exclusive use of e-cigarettes containing nicotine (mean concentration, 28 ± 14 mg/mL) and flavorings. More than 10 flavors were reported, with common categories including mint and fruit variations. Fifteen participants with e-cigarette exposure reported using closed-system, pod-style disposable fourth-generation e-cigarette devices, which use nicotine salts; the remaining 5 reported using pod-style third-generation devices, which use freebase e-liquid and are highly customizable.[33]

Exercise Capacity

Individual $\dot{V}O_{2peak}$ values and estimated group means are displayed in Figure 1A. Independent of age, height, and sex, participants with e-cigarette exposure exhibited a lower $\dot{V}O_{2peak}$ compared with control participants (estimated mean ± SE $\dot{V}O_{2peak}$, participants with e-cigarette exposure, 29.6 ± 1.5 mL/kg per minute vs control participants, 34.8 ± 1.5 mL/kg per minute; $P_{group} = .017$) (Table 2). Unadjusted values and 95% CIs are presented in e-Tables 1 and 2.

Ventilatory Responses to Exercise

Individual $\dot{V}_E/\dot{V}CO_2$ nadir values and estimated group means are displayed in Figure 1B. Independent of age,

TABLE 1] Participant Characteristics (N = 40)

Characteristic	Control	Particpants With E-Cigarette Exposure	*P* Value
Male/female, No.	12/8	12/8	
Age, y	24 ± 4	23 ± 4	.631
Height, cm	176 ± 11	173 ± 11	.391
Weight, kg	74 ± 18	78 ± 18	.505
BMI, kg/m^2	24 ± 4	26 ± 5	.153
E-cigarette puff-years	**0 ± 0**	**3.4 ± 2.9**	**< .001**
Weekly MVPA, min	175 ± 111	145 ± 126	.429
Pulmonary function			
FEV_1, % predicted	98.4 ± 18.2	102.7 ± 13.5	.406
FEV_1, L	4.0 ± 1.1	4.0 ± 0.8	.989
FVC, % predicted	103.5 ± 19.0	112.6 ± 13.5	.096
FVC, L	4.9 ± 1.4	5.1 ± 1.2	.662
FEV_1/FVC, % predicted	95.5 (6.0)	94.5 (9.5)	.360
FEV_1/FVC, ratio	0.82 (0.05)	0.83 (0.11)	.877
TLC, % predicted	102.1 ± 17.4	97.5 ± 10.5	.442
IC, % predicted	97.9 ± 23.4	108.4 ± 17.5	.127
D_{LCO}, % predicted	100.7 ± 17.5	105.8 ± 10.5	.276
D_{LCO}, mL/min/mm Hg	30.1± 8.6	30.8± 7.1	.765
V_A, % predicted	95.8 ± 18.5	101.2 ± 14.1	.313
V_C, mL	77.6 ± 22.8	93.2 ± 27.0	.915
D_M, mL/min/mm Hg	72.2 ± 56.2	87.3 ± 75.0	.349

Data are presented as absolute means ± SD or median (interquartile range) unless otherwise noted. Bold indicates significant difference ($P < .05$) between control vs e-cigarette. D_{LCO} = pulmonary diffusing capacity for carbon monoxide; D_M = diffusing membrane capacity; IC = inspiratory capacity; MVPA = minutes of moderate to vigorous physical activity; TLC = total lung capacity; V_A = alveolar volume; V_C = pulmonary capillary blood volume.

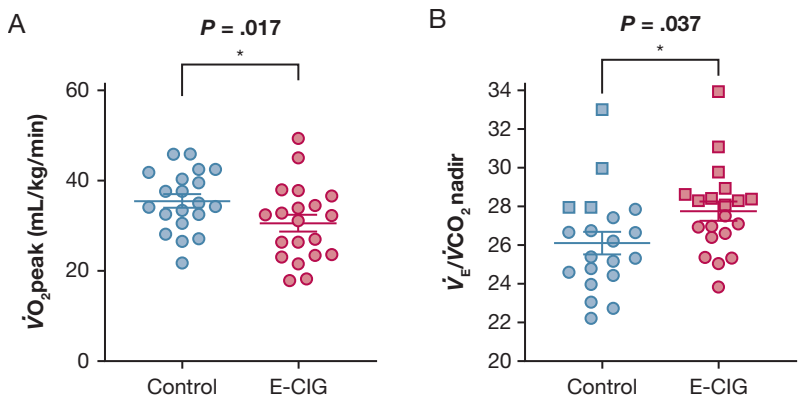


Figure 1 – *A, Exercise capacity, as evaluated by peak rate of oxygen consumption relative to body mass ($\dot{V}O_2$, mL/kg/min) in E-CIG and control participants. B, Ventilatory efficiency, represented by the ventilatory equivalent for CO_2 production ($\dot{V}_E/\dot{V}CO_2$ nadir), during exercise in E-CIG and control participants. Square symbols denote $\dot{V}_E/\dot{V}CO_2$ nadir values exceeding predicted upper limit of normal. Data are presented as individual values and marginal means ± SE, adjusted for age, height, sex, and fitness. n = 20 per group. Bold indicates significant difference (P < .05). E-CIG = participants with e-cigarette exposure; $\dot{V}_E$ = minute ventilation; $\dot{V}CO_2$ = CO_2 output; $\dot{V}O_{2peak}$ = peak rate of oxygen consumption.*

height, and sex, participants with e-cigarette exposure exhibited an elevated $\dot{V}_E/\dot{V}CO_2$ nadir during submaximal exercise (estimated mean ± SE $\dot{V}_E/\dot{V}CO_2$ nadir, participants with e-cigarette exposure, 27.91 ± 0.45 vs control participants, 26.40 ± 0.44; P_{group} = .037). Ten of 20 participants with e-cigarette exposure exhibited an impaired $\dot{V}_E/\dot{V}CO_2$ nadir above their individual predicted upper limit of normal compared with 4 of 20 control participants. End-tidal CO_2 did not differ between groups at the workload corresponding to each participant's respective $\dot{V}_E/\dot{V}CO_2$ nadir (participants with e-cigarette exposure, 39.5 ± 0.8 vs control participants, 41.4 ± 0.8 mm Hg; P_{group} = .074).

TABLE 2] Select Cardiopulmonary and Perceptual Responses to Peak Exercise in Control Participants and Participants With E-Cigarette Exposure

Peak Data	Control	Particpants With E-Cigarette Exposure	*P* Value
$\dot{V}O_2$, L/min	2.5 ± 0.1	2.3 ± 0.1	.229
$\dot{V}O_2$, mL/kg/min	**34.8 ± 1.5**	**29.6 ± 1.5**	**.017**
$\dot{V}O_2$, % predicted	106 ± 4	95 ± 4	.066
Power, W	192 ± 9	170 ± 10	.091
Power, % predicted	102 ± 5	89 ± 5	.056
Respiratory Quotient	1.3 ± 0.04	1.3 ± 0.04	.512
Heart rate, beats/min	186 ± 3	181 ± 3	.244
Heart rate, % predicted	98 ± 2	94 ± 2	.125
Ventilation, L/min	101 ± 5	96 ± 5	.503
Ventilation, % predicted	98 ± 5	92 ± 5	.402
f_B, breaths/min	42 ± 2	42 ± 2	.891
V_T, L	2.4 ± 0.1	2.4 ± 0.1	.857
Spo_2, %	98 ± 1	97 ± 1	.891
Dyspnea, 0-10 Borg	6.8 ± 0.4	7.7 ± 0.5	.202
Dyspnea/$\dot{V}_E$, L/min	0.08 ± 0.01	0.09 ± 0.01	.225
Dyspnea/$\dot{V}O_2$, L/min	2.9 ± 0.3	3.5 ± 0.3	.093
Dyspnea/workload, W	**0.04 ± 0.01**	**0.05 ± 0.02**	**.042**
Leg discomfort, 0-10 Borg	**8.8 ± 0.4**	**7.2 ± 0.4**	**.005**
Leg discomfort/$\dot{V}O_2$, L/min	3.8 ± 0.2	3.3 ± 0.2	.142
Leg discomfort/workload, W	0.05 ± 0.01	0.05 ± 0.01	.811
($EILV_{\%TLC}$)/$\dot{V}_E$, L/min	0.98 ± 0.07	1.06 ± 0.07	.470

Data are presented as estimated marginal means ± SE, adjusted for age, height, and sex. Bold indicates significant difference ($P < .05$) between control participants vs participants with e-cigarette exposure. $EILV_{\%TLC}$ = end-inspiratory lung volume as percent total lung capacity; f_B = frequency of breaths; Spo_2 = oxygen saturation; $\dot{V}_E$ = minute ventilation; $\dot{V}O_2$ = rate of oxygen consumption; V_T = tidal volume.

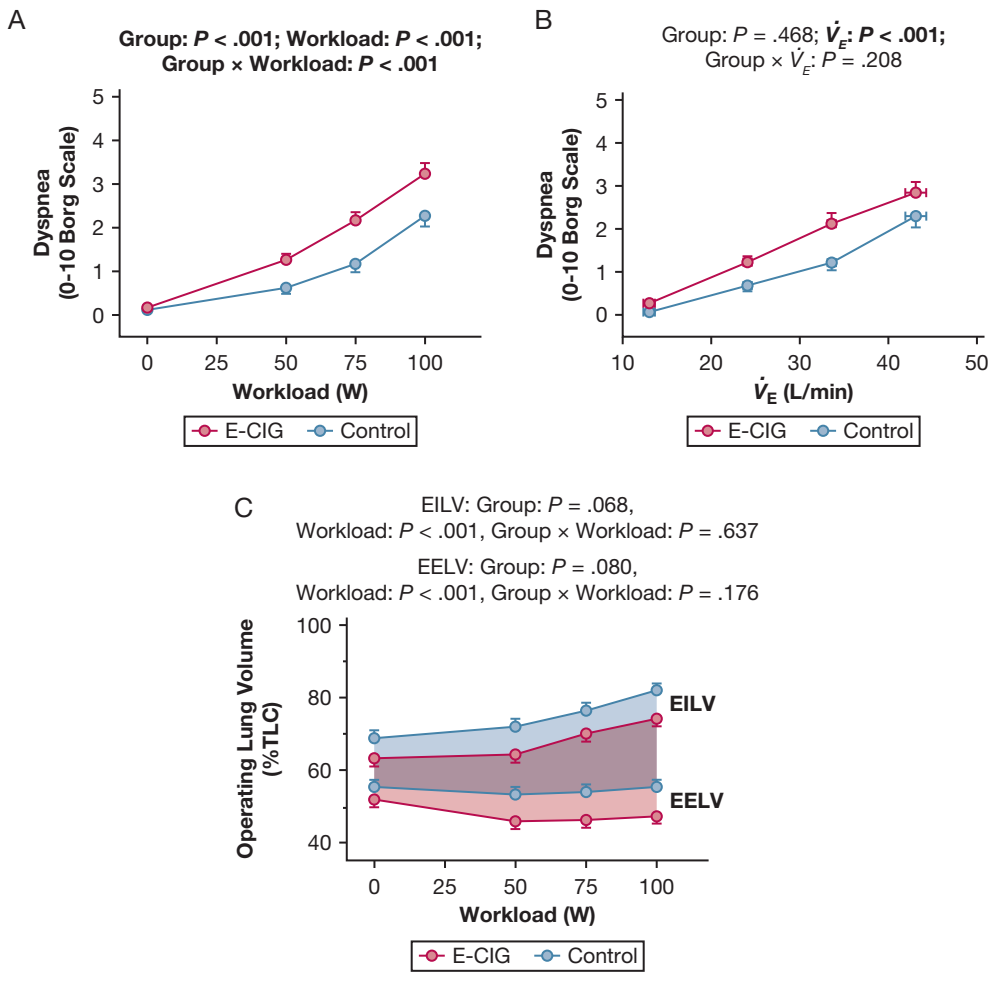


Figure 2 – *Exertional dyspnea responses to submaximal exercise relative to workload (A), exertional dyspnea responses to submaximal exercise relative to* $\dot{V}_E$ *(B), and operating lung volume responses (C) during submaximal exercise in E-CIG and control participants. Data are expressed as marginal means ± SE, adjusted for age, height, sex, and fitness. n = 20 per group. Bold indicates significant difference (P < .05). E-CIG = participants with e-cigarette exposure;* $EILV_{\%TLC}$ *= end-inspiratory lung volume as a percentage of total lung capacity;* $EELV_{\%TLC}$ *= end-expiratory lung volume as a percentage of total lung capacity;* $\dot{V}_E$ *= minute ventilation.*

No between-group differences were detected in operating lung volumes throughout submaximal exercise ($EILV_{\%TLC}$, P_{group} = .068; $EELV_{\%TLC}$, P_{group} = .080), and the rate of change in operating lung volumes with increasing workload did not differ between groups ($EILV_{\%TLC}$, $P_{group*workload}$ = .637; $EELV_{\%TLC}$, $P_{group*workload}$ = .176) (Fig 2). End-exercise $EILV_{\%TLC}$ as a function of $\dot{V}_E$ did not differ between groups (P_{group} = .470), and the prevalence of dynamic hyperinflation (decreased inspiratory capacity from rest-to-peak > 150 mL) did not differ between groups (participants with e-cigarette exposure, n = 2; control participants, n = 4; P_{group} = .419).

Dyspnea Response

Independent of age, height, sex, and $\dot{V}O_{2peak}$, participants with e-cigarette exposure reported greater dyspnea during submaximal exercise, as well as a greater rise in dyspnea with increasing workload, compared with control participants (P_{group} < .001, $P_{group*workload}$ < .001) (Fig 3). When analyzed as a function of $\dot{V}_E$, dyspnea responses did not differ between groups (P_{group} = .468), and the rise in dyspnea as a function of $\dot{V}_E$ was not different between groups ($P_{group*}\dot{V}_E$ = .208).

Dyspnea at peak exercise did not differ between groups when analyzed as absolute ratings, or relative to peak workload, or $\dot{V}_E$, independent of age, height, and sex (all, P > .05) (Table 2).

Postural D_{LCO} Recruitment

Results from LME models evaluating between-group differences in D_{LCO}, V_C, and D_M in response to postural

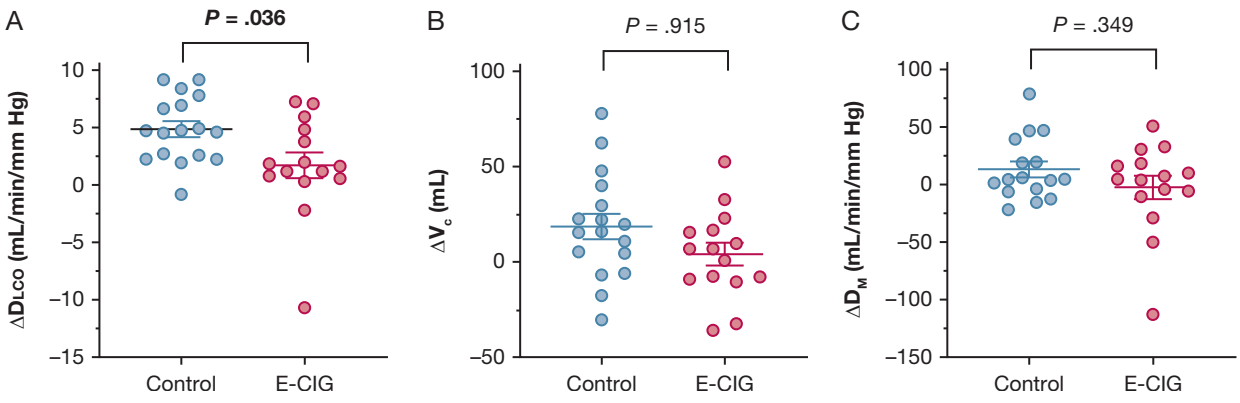


Figure 3 – *Change (Δ) in DLCO, V_C, and D_M from the upright to supine posture in e-cigarette vs control groups. Data are presented as individual values and marginal means ± SE, adjusted for age, height, sex, and fitness. E-CIGs, n = 16; control, n = 17. Bold indicates significant difference (P < .05). DLCO = pulmonary diffusing capacity for carbon monoxide; D_M = pulmonary membrane diffusing capacity; E-CIG = participants with e-cigarette exposure; V_C = pulmonary capillary blood volume.*

changes are provided. Participants with e-cigarette exposure displayed a blunted postural DLCO recruitment ($P_{group*posture}$ = .036) (Fig 3A), independent of age, height, sex, and $\dot{V}O_{2peak}$. Postural changes in V_C and D_M did not differ between groups (both $P_{group*posture} \geq$ .349) (Fig 3B, 3C). A sensitivity analysis replaced absolute DLCO with relative change of DLCO, and results remained unchanged.

Discussion

This study reveals that otherwise healthy, young adults who report regular e-cigarette use with normal pulmonary function exhibit significant cardiopulmonary abnormalities during exercise, as well as impaired pulmonary vascular responsiveness to a hemodynamic postural stress. The main findings of this investigation were that participants with chronic e-cigarette exposure exhibited: (1) reduced exercise capacity; (2) impaired ventilatory efficiency during exercise; (3) exaggerated dyspnea response to exercise; and (4) blunted recruitment of diffusing capacity in response to postural change. These results suggest that young adults who report regular e-cigarette use exhibit cardiopulmonary and pulmonary vascular abnormalities, which may precede more overt impairment with continued exposure.

Reduced Exercise Capacity

Mean $\dot{V}O_{2peak}$ was 5.2 ± 2.1 mL/kg per minute lower among participants with chronic e-cigarette exposure compared with control participants (95% CI, 1.0-9.5). $\dot{V}O_{2peak}$ is an independent predictor of all-cause mortality.[34] Among healthy older male individuals, a 3.5 mL/kg per minute decrement in $\dot{V}O_{2peak}$ is associated with a 19% increase in mortality. Assuming similar associations among young adults, the observed 5.2 mL/kg per minute reduction in $\dot{V}O_{2peak}$ displayed by participants with e-cigarette exposure may be clinically significant.

Self-reported physical activity did not differ between groups (Table 1), suggesting that reduced exercise capacity among participants with e-cigarette exposure is likely not attributable to differences in physical activity levels. This finding extends previous research citing reduced exercise capacity in this population[14] and suggests that important physiological abnormalities, independent of physical activity, may contribute to a reduced exercise capacity among individuals with chronic e-cigarette exposure.

Impaired Ventilatory Efficiency

Ventilatory efficiency is represented by the ratio of ventilation to CO_2 production ($\dot{V}_E/\dot{V}CO_2$) and is typically quantified as the nadir value obtained during a CPET.[26] Impaired ventilatory efficiency is a hallmark of chronic respiratory and cardiovascular conditions,[24] is associated with reduced exercise capacity and greater exertional dyspnea, and is predictive of mortality in various cardiopulmonary conditions.[35-37] Participants with e-cigarette exposure exhibited a decreased ventilatory efficiency, with 50% of displaying an impaired ventilatory efficiency above their predicted upper limit of normal.[38] Ventilatory inefficiency can be attributed to alveolar hyperventilation and/or elevated physiological dead space.[26] End-tidal CO_2 at the power output corresponding to the $\dot{V}_E/\dot{V}CO_2$ nadir was not different between participants with e-cigarette exposure and control participants, suggesting that the decreased

ventilatory efficiency observed among participants with e-cigarette exposure is not likely explained by hyperventilation. Accordingly, the elevated $\dot{V}_E/\dot{V}CO_2$ nadir in participants with e-cigarette exposure may reflect increased physiological dead space.[26]

Previous research in mild COPD suggests that early pulmonary vascular dysfunction contributes to hypoperfusion, increased dead space ventilation, elevated $\dot{V}_E/\dot{V}CO_2$, exaggerated exertional dyspnea, and reduced exercise capacity.[25,39,40] Thus, the elevated $\dot{V}_E/\dot{V}CO_2$ nadir observed in participants with e-cigarette exposure may reflect physiological dead space in the presence of impaired pulmonary perfusion, similar to the pathophysiology observed in mild COPD.[41] Further investigation using arterial blood sample analysis to directly quantify physiological dead space is warranted to elucidate the underlying physiology in individuals with chronic e-cigarette exposure.

Exaggerated Dyspnea Response to Exercise

Participants with e-cigarette exposure showed exaggerated exertional dyspnea compared with control participants, which exceeded the minimal clinically important difference of 1 Borg unit at the highest equivalent work rate of 100 W.[42] The physiological mechanisms underlying an exaggerated exertional dyspnea may be attributed to altered breathing mechanics and/or abnormally high ventilatory demand.[13] To elucidate the primary contributing mechanisms, dyspnea was analyzed relative to both workload and ventilation. Our results indicate that participants with e-cigarette exposure showed a greater increase in dyspnea when analyzed relative to workload; however, dyspnea was not different between groups when analyzed relative to ventilation. This pattern suggests that participants with e-cigarette exposure sense dyspnea appropriately relative to a given ventilatory demand (Fig 2).[13] An abnormal dyspnea response observed relative to workload may be attributed to altered breathing mechanics or a greater ventilatory demand. Operating lung volume data indicate that participants with e-cigarette exposure did not develop dynamic hyperinflation, suggesting the abnormal dyspnea response to submaximal exercise was likely not explained by altered breathing mechanics, and therefore the abnormal dyspnea may be attributed to a disproportionate ventilatory demand. Indeed, a disproportionate ventilatory demand would be consistent with the elevated $\dot{V}_E/\dot{V}CO_2$ nadir observed among participants with e-cigarette exposure as described earlier.

Blunted D_{LCO} Recruitment

In the upright seated posture, gravity creates regional differences in pulmonary perfusion,[35] which are abolished in the supine posture through the reduction of the hydrostatic column and a central shift in blood volume.[43] In turn, these effects lead to capillary recruitment,[44] altogether increasing D_{LCO}.[17] Participants with e-cigarette exposure exhibited a blunted D_{LCO} recruitment, a pattern suggestive of abnormal pulmonary vascular responsiveness.[16,17] Postural recruitment of D_{LCO} in the supine position is reduced with aging[17] and further impaired among individuals with mild COPD,[16] a population exhibiting pulmonary vascular dysfunction and destruction.[12] Research supports the hypothesis that young adults with chronic e-cigarette exposure may manifest pulmonary vascular abnormalities, as impaired ventilation/perfusion matching in these participants was found to occur alongside disrupted pulmonary vascular perfusion.[9] Furthermore, previous preclinical and human studies have found that e-cigarette use may impair systemic vascular function.[10,45,46] Altogether, these results suggest a pattern of compromised vascular function/structure in young adults with chronic e-cigarette exposure.

Study Limitations

This study has several limitations. The study provides a deep physiological characterization of a relatively small sample size. Physical activity was assessed using a validated self-report questionnaire[18] but not directly measured through gold standard activity monitors, potentially introducing bias and error in physical activity quantification. The value of the rate of carbon monoxide uptake by RBCs used in the multiple F_{IO_2} technique was assumed to be the same across participants. Because carboxyhemoglobin levels were low and within normal range among participants with e-cigarette exposure, we suggest this assumption is appropriate. Participants reported using $>$ 10 different e-cigarette flavorings, limiting the ability to stratify by flavoring in a small sample. Future larger trials should examine the influence of flavorings and exposure time on cardiopulmonary outcomes. E-cigarette, tobacco cigarette, and cannabis exposure was determined by using self-reported tools, which are limited in providing the most accurate quantification of exposure. Finally, it is unclear how e-cigarette responses compare to those of traditional tobacco smoking or a mix of e-cigarette and tobacco exposure.

Interpretation

The current findings challenge the assumption that e-cigarette use is a safe alternative to tobacco cigarette use.[2] Among ostensibly healthy, young adults who report regular e-cigarette use and present with normal resting pulmonary function, significant cardiopulmonary abnormalities were observed during exercise in addition to blunted pulmonary vascular responsiveness. Over time, these subclinical impairments among young adults with chronic e-cigarette use may evolve to overt cardiopulmonary dysfunction.

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